

Junhyeong Lee

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EDUCATION

May. 2021 ~ Present	Sangmyung University Department of Software	Republic of Korea
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Bachelor Student
GPA: 4.01 / 4.5

RESEARCH INTERESTS

- Spatio-Temporal Video Representation Learning
- Vision-Language Models (VLM) & Vision-Language-Action (VLA)
- Domain Generalization & Multimodal Reasoning

PUBLICATIONS

<SCIE>

1. **Junhyeong Lee**, Hansaem Park, "Spatio-Temporal Representation Learning via Unified VLA Backbone for Video Frame Interpolation," Target: *IEEE Access* (In Preparation)

<DOMESTIC>

1. **Junhyeong Lee**, Hansaem Park, "Real-Time KBO Check Swing Automated Adjudication System: YOLO11 Keypoint Detection and Threshold Calibration Approach," *Journal of Korea Multimedia Society*, vol. 29, no. 7, pp. 1026-1035, 2026.

CONFERENCES

1. Minseok Kwon, Dongwoo Kim, **Junhyeong Lee**, Hansaem Park, "A Study on A Channel-Independent and Channel-Mixing Hybrid Model for Enhanced OOD Adaptability in Building Energy Forecasting," Proceedings of the KICS Summer Conference 2026, Jeju, Korea, Jun. 2026. (Oral presentation)

RESEARCH & WORK EXPERIENCES

- Research Student at AMI LAB, *Sangmyung University (Jan. 2025 ~ Present)*
- Python Programming Tutor *Coderland (Mar. 2024 ~ Present)*

PROJECTS

- **Spatial Foundation Model for 3D Indoor Microclimate Prediction** *Feb. 2026 – Present*
 - **Affiliation:** Ministry of Science and ICT (MSIT)
 - **Description:** Combined a multimodal SFM encoder integrating spatial, facility, and sensor data with a Physics-Informed CFD-Surrogate model to predict 3D indoor microclimate (temperature, airflow, CO₂).
 - **Role:** Designed the multimodal SFM encoder architecture; conducted zero-shot generalization evaluation on unseen conditions and risk-scenario analysis.

- EnergyOpt: LLM-Agent-Based Energy Analytics System** Oct. 2025 – Dec. 2025
 - Affiliation:** Sangmyung University Tech Entrepreneurship Club
 - Description:** Designed an AI-agent-based energy analytics system that auto-generates and executes analysis code from natural language queries; contributed to a patent specification draft.
 - Role:** Performed statistical and pattern analysis (usage patterns, peak-demand intervals) to support the AI-agent's decision-making.
- Real-Time KBO Check-Swing Automated Adjudication System** Sep. 2025 – Apr. 2026
 - Affiliation:** Individual Research (Academic Paper Project)
 - Description:** Researched a real-time deep learning adjudication pipeline to determine check-swings by compensating for perspective distortion in single broadcast feeds.
 - Role:** Built a dual-model framework based on YOLO11n-pose, designed geometric correction thresholds, and implemented a 2-frame temporal consistency verification logic (Published as 1st Author in Korea Multimedia Society).
- Autonomous 1/10-Scale ASW Racer Navigation System** May. 2025 – Nov. 2025
 - Affiliation:** Korea Institute for Robot Industry Advancement (KIRIA) -International Robot Contest
 - Description:** Developed a real-time lane tracking and obstacle avoidance system by integrating an OpenCV perception pipeline into the ROS control loop.
 - Role:** Managed the OpenCV-based image processing perception pipeline and integrated ROS mobility packages.
- ROK Marine Corps Integrated Administrative System SW Overhaul** Feb. 2023 – Apr. 2023
 - Affiliation:** Republic of Korea Marine Corps Headquarters
 - Description:** Restructured and modernized the military administrative log logging architecture and system infrastructure.
 - Role:** Developed and optimized front-end interfaces using HTML and CSS.

AWARDS & SCHOLARSHIPS

- Encouragement Award, Capstone Design Competition (AI Track) Sangmyung University (Aug. 2026)
- 2nd Place (Intellectual Property Commissioner's Award) KIRIA (Nov. 2025)
- Lieutenant General's Citation (SW Development Excellence) ROK Marine Corps (Jul. 2023)
- Study Scholarship (Academic Excellence) Sangmyung University (Dec. 2024)
- Study Scholarship (Academic Excellence) Sangmyung University (Dec. 2021)
- SW Human Resources Scholarship Sangmyung University (Jun. 2021)

CERTIFICATIONS & EDUCATION PROGRAMS

- TOEIC 695 (Jun. 2026)
- Advanced Data Analytics Semi-Professional (ADsP) Korea Data Agency (KDATA) (May. 2026)
- LG AIMERS (9th Cohort) LG AI Research (Jul. 2026 ~ Present)
- Google AI Essentials Google (via Coursera) (Sep. 2025)

SKILLS & TECHNIQUES

- AI & Vision Frameworks: PyTorch, OpenCV
- Programming Languages: Python, C, C++, SQL, Java, HTML, CSS
- Environments & Tools: ROS, Linux, Git